

Activity 10 Pre-Lab — Crossed Aldol (2-Acetylpyridine + 4-Nitrobenzaldehyde)

[Your name] & [partner]

[MM/DD/YY]

Organic Lab · aldol addition (RT) vs. condensation (50 °C)

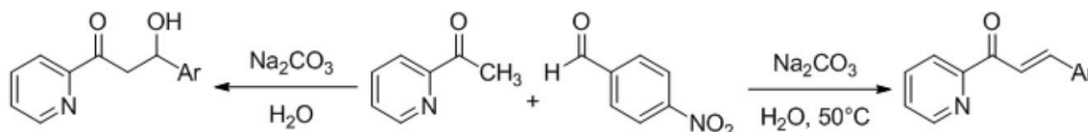
■ Black = write this in your notebook ■ Blue = context, don't copy it

Copy the header onto every page. Your pair does Part 1 or Part 2 — write the one you're assigned.

Purpose

Run a base-catalyzed **crossed aldol** of **2-acetylpyridine** (enolate) with **4-nitrobenzaldehyde** (electrophile) and show that **conditions decide the product**: **room temp** → **aldol addition** (β -hydroxy ketone) vs **50 °C** → **aldol condensation** (α,β -unsaturated ketone + H_2O).

🔥 DRAW — both outcomes (from the procedure)



Na_2CO_3

makes the enolate at the acetyl CH_3 → attacks the aldehyde. RT stops at the β -hydroxy ketone (addition); 50 °C dehydrates to the conjugated enone (condensation).

Chemical Reagents & Safety Table

CONTEXT ▶ Required: 2-acetylpyridine, 4-nitrobenzaldehyde, sodium carbonate, methanol — plus one row per product (MW only, per directions).

Chemical	Role	MW	Amount	Physical properties	GHS pictograms
2-Acetylpyridine	Enolate (nucleophile)	121.14	~0.2 mL (weigh exact)	Yellow liquid; bp 192 °C; d 1.08 g/mL	 irritant
4-Nitrobenzaldehyde	Aldehyde (electrophile)	151.12	300 mg	Pale-yellow solid; mp 106 °C	 irritant
Sodium carbonate (0.54% w/v)	Base (makes enolate)	105.99	10 mL	Colorless aqueous solution	 irritant
Methanol	Dissolves the aldehyde	32.04	5–7 mL	Colorless liquid; bp 65 °C; d 0.79 g/mL; flash pt 11 °C	   flammable toxic health
Addition product (β -hydroxy ketone)	Product — Part 1 (RT)	272.26	— (MW only)	—	—
Condensation product (enone)	Product — Part 2 (50 °C)	254.24	— (MW only)	—	—

FYI ▶ condensation MW = addition MW – 18 (loses one H_2O).

GRAB FIRST: 100-mL RBF + stir bar · disposable 1-mL syringe · **cork ring** (to weigh the flask) · 25-mL Erlenmeyer · steam bath · funnel · stir plate on a **raised lab jack** · Büchner + filter flask + vacuum · filter paper. **Part 2 also:** blue IKA stir plate w/ **heating block** · condenser + 2 hoses · **temperature sensor** + small 3-prong clamp.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Set-up (same for Part 1 and Part 2)

1. Tare the balance with a **cork ring** on it. Put a **stir bar + the 1-mL syringe** into a **100-mL RBF**; **weigh the whole assembly** and record.
2. At the dispensing station, use that same syringe to withdraw **~0.2 mL 2-acetylpyridine**; **recap the reagent bottle**; put the loaded syringe back in the flask; re-zero the balance w/ a cork ring and **reweigh** → record. **Mass of 2-acetylpyridine = difference.**
3. Squirt the 2-acetylpyridine from the syringe into the RBF, then add **20 mL H₂O**. **Later, rinse the syringe 2× w/ acetone into the waste beaker** → syringe to solid waste.
4. In a **25-mL Erlenmeyer**: **300 mg 4-nitrobenzaldehyde + 5 mL MeOH**. Swirl over a **steam bath** to dissolve. **If solid remains, add 2 mL more MeOH and keep warming.**
5. **While still warm**, add that MeOH solution to the RBF **through a funnel**. **If it cools first the aldehyde crashes out.**
6. Add **10 mL 0.54% Na₂CO₃** through the same funnel.

Part 1 — Addition (room temp)

7. Put a stir plate on a **raised lab jack**; clamp the flask above it; stir **vigorously for 60 min at room temperature.**

Part 2 — Condensation (45–50 °C)

7. Put the blue **IKA stir plate + heating block** on a raised lab jack; clamp the flask on the block. **Jack must lower far enough that the block no longer touches the flask.**
8. Attach condenser hoses: **water IN the bottom nozzle, drain OUT the top**; drain hose secured in the cup sink. One partner **holds the hoses on** while the other turns water to a strong stream, then dials it back to a **steady drip**. **Strong flow first seats a hidden valve; holding the hoses stops them popping off and flooding the hood.**
9. Insert the **temperature sensor through the condenser** into the solution — **submerged, not touching the stir bar or flask bottom**. Clamp its rubber connector with a small 3-prong clamp.
10. **Instructor inspects the set-up.** Power on, stir, set **45–50 °C**. Once the sensor reads 45–50 °C, stir **60 min**.
11. **Never exceed 50 °C** — overheating gives brown goo (side reactions), and the flask is half full so you don't want reflux.

Both parts — isolate

12. After 60 min, collect the solid by **vacuum filtration**. Wash with **DI H₂O (2 × 20 mL)**. Keep pulling vacuum **10 min**.
13. Scrape the solid onto **dry filter paper**; label it **ADDITION** or **CONDENSATION** product.
14. Once dry, **weigh** → % yield. Dissolve a little in EtOAc and run a **TLC (30% EtOAc/hexanes)**, spotting your product beside authentic **2-acetylpyridine** and **4-nitrobenzaldehyde**.
15. Leave the product to dry in your drawer — **IR next week**. Share data with the complementary group.

✍ WRITE

Experiment Notes (in lab)

Which part: _____ (RT = addition / 50 °C = condensation)

measurement	value
Mass assembly (flask+bar+syringe)	g
Mass after 2-acetylpyridine	g
Mass 2-acetylpyridine	g
Reaction temp held	°C
Product mass (dry)	g
% yield	

Appearance during stir (solid forming? color):

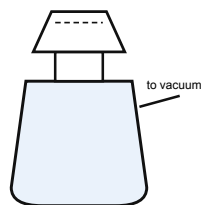
TLC R_f

lane	R _f
your product	
2-acetylpyridine	
4-nitrobenzaldehyde	

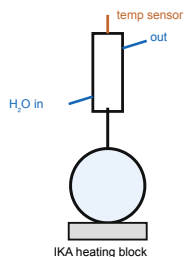
Other group's product (compare):

limiting reagent: compare mol 2-acetylpyridine ($\text{mass} \div 121.14$) vs 4-nitrobenzaldehyde ($0.300 \text{ g} \div 151.12$) $\rightarrow$ theoretical product $\rightarrow$ % yield.

Setup — draw this



vacuum filtration



Part 2: condenser + sensor

CONTEXT (don't copy) ▶

Addition vs condensation: the enolate adds to the aldehyde giving a β -hydroxy ketone. Heat (plus the electron-poor nitro ring) drives loss of water $\rightarrow$ conjugated enone. Same reagents; temperature picks the product.

Why it precipitates: the nitro product is insoluble in the water/MeOH mix $\rightarrow$ filters straight out.

Clean-up & Conclusions

- Filtrates ($\text{Na}_2\text{CO}_3 + \text{MeOH}$) $\rightarrow$ **aqueous waste**; excess starting materials & acetone/TLC solvent $\rightarrow$ **organic waste**; excess 4-nitrobenzaldehyde $\rightarrow$ solid waste; bench task.
- Conclusions** (~4 sentences [after lab](#)): which product your conditions gave & why (RT vs 50 °C); % yield; how TLC/mp/IR distinguish addition (O–H, $\text{sp}^3 \text{C–H}$) from condensation (C=C , conjugation); comparison with the other group.